

Nandha R

Data Scientist & AI / ML Engineer

Chennai (Current) & Dharmapuri (Native), Tamil Nadu
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Objective: To work in an environment offering challenging career opportunities where I can apply and enhance my skills for solving diverse technical problems in Data Science, Machine Learning, and AI Systems.

TECHNICAL SKILLS

Programming Languages: Python, Java, SQL

AI & Machine Learning: AutoML, XGBoost, LightGBM, CatBoost, Scikit-learn, RAG, ChromaDB, Ollama, Predictive Modeling, Regression, Classification, Feature Engineering, Hyperparameter Tuning, Benchmarking, EDA

NLP & Speech AI: Speech Recognition (SR), Natural Language Processing (NLP), Text-to-Speech (TTS), Sentence Transformers, BM25 Hybrid Search

Frameworks & Tools: FastAPI, React.js, Streamlit, Docker, Python Data Pipelines

EXPERIENCE

Data Scientist Intern — Gradtwin (Chennai, Tamil Nadu) June 2026 – August 2026

- Architected end-to-end Python data pipelines and performed rigorous EDA on multi-source datasets to uncover actionable business insights for client use cases.
- Developed and optimized predictive modeling workflows—including regression and classification pipelines—improving forecasting accuracy through systematic feature engineering, hyperparameter tuning, and model benchmarking.
- Designed and deployed a production-grade RAG AI chatbot, integrating LLM APIs, ChromaDB vector storage, and semantic document retrieval to enhance operational workflows.
- Cleaned, transformed, and engineered features from complex, high-volume datasets, significantly accelerating data preprocessing pipelines.
- Collaborated cross-functionally with engineering and product teams to implement ML-driven solutions that directly improved operational efficiency.

FEATURED PROJECTS

AI-Powered Data Cleaning and Imputation Recommendation Engine (DataClean AI) Aug 2026

Stack: FastAPI, React.js, Scikit-learn, XGBoost, LightGBM, CatBoost, Docker.

- Automated Quality Detection:** Autonomously detects 13+ data anomalies including missingness patterns (MCAR/MAR), multivariate outliers (IQR/Z-score/MAD), distribution skewness, and class imbalance.
- Hybrid AI Recommendation:** Combines 30+ domain heuristic rules with a Voting Ensemble classifier (XGBoost + LightGBM + CatBoost) achieving 97.8% accuracy across 25 cleaning techniques.
- Scalable Streaming Engine:** 10MB chunked multipart upload pipeline handling multi-gigabyte datasets (3M+ rows) with zero-OOM memory streaming and automated PDF audit reporting.

PDF Summarizer Chatbot using RAG (PaperBrain) Jul 2026

Stack: FastAPI, Streamlit, ChromaDB, Sentence Transformers, BM25, Ollama (Local LLMs).

- Local & Privacy-First:** Fully offline RAG system where embeddings, vector storage, and language generation run locally without data leaving the machine.
- Hybrid Search:** Fuses dense vector similarity search (Sentence Transformers) with BM25 keyword matching for context retrieval and document summarization.

The Voice Assistant with Desktop Accessibility Mar 2026

Stack: Python, Speech Recognition (SR), NLP Intent Engine, Text-to-Speech (TTS), OS Automation.

- Hands-free desktop voice assistant with Speech Recognition, NLP command parsing, and responsive TTS audio feedback for seamless desktop navigation.

Residential Property Valuation Using Machine Learning Feb 2025

Stack: Python, Scikit-Learn, Pandas, NumPy, Matplotlib, Seaborn.

- Engineered regression modeling workflows on large real estate datasets to accurately forecast property market valuations based on location, area, and amenities.

EDUCATION

Bachelor of Engineering in Computer Science and Engineering (BE CSE) 73.76%
Vinayaka Mission's Kirupananda Variyar Engineering College, Salem, Tamil Nadu (2022 – 2026)

Higher Secondary School Examination (HSC) 52.8%
Government Higher Secondary School, Salem, Tamil Nadu (2021 – 2022)

Secondary School Examination (SSLC) 72.5%
Government Higher Secondary School, Salem, Tamil Nadu (2019 – 2020)

CERTIFICATES & ACHIEVEMENTS

1. **GenAI Powered Data Analytics Job Simulation (Jan 2026)** — Completed Tata (via Forage) simulation in EDA, AI risk profiling, and delinquency prediction. (Verification: *SFPHHSaWj4nHTiKK7*)

2. **1st Prize Winner – Kongu Engineering College IPL AUCTION 2025** — Analytics, valuation, and bidding strategy optimization.
3. **Data Science in Python Training Program (2024)** — IIIT KOTTAYAM Engineering College intensive training in Python data science.
4. **Green Skill and AI Course Certification (2025)** — Edunet Foundation course on AI fundamentals and sustainable technologies.
5. **Paper Presentation on 'IOT in Military Security' (2023)** — KSR Engineering College technical presentation.